Customer Behavior Analysis

A structured retail/e-commerce dataset used to analyze purchasing patterns, optimize marketing, and drive business decisions through Python, SQL, and Power BI.

PYTHON · SQL · POWER BI

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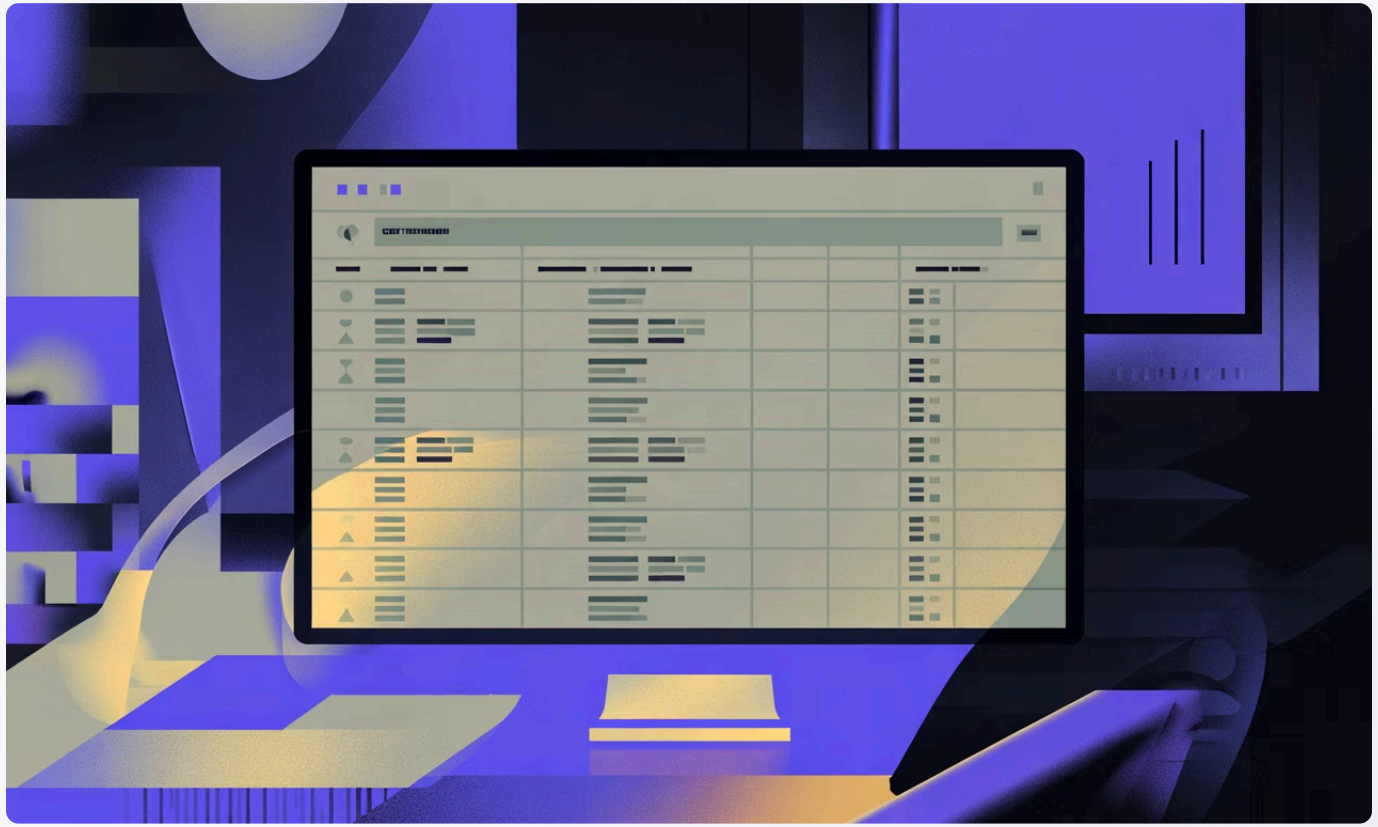
About the Dataset



Dataset Profile

- **Type:** Retail / E-commerce
- **Structure:** Tabular, Customer Transaction Level
- **Numerical:** Age, Purchase Amount, Previous Purchases
- **Categorical:** Category, Payment Method, Season

Why It Matters

- Identify high-value customers
- Understand purchase patterns
- Optimize discount strategies
- Increase Customer Lifetime Value (CLV)

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The Business Problem

How can a retail company understand customer purchasing behavior to **increase revenue, improve retention, and optimize marketing?**

Low Retention

Customers not purchasing frequently — need to identify frequent vs. one-time buyers

Ineffective Campaigns

Do discounts increase purchases? Which customers respond to promo codes?

Product Demand

Which categories perform best? Which products generate the highest revenue?

Segmentation

Identify high-value, frequent, and discount-sensitive customer groups

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Data Dictionary

Key columns in the dataset covering customer identity, product details, transaction values, and behavioral signals.

Customer ID	Unique identifier for each customer
Age / Gender	Customer demographics
Item Purchased / Category	Product and category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent per transaction
Season / Location	Purchase season and geographic location
Review Rating	Customer product rating
Subscription Status	Whether customer is subscribed (Yes/No)
Discount Applied / Promo Code	Discount and promo usage flags
Previous Purchases	Number of past purchases
Payment Method	Mode of payment
Shipping Type	Type of shipping selected
Frequency of Purchases	Purchase frequency (Weekly, Monthly, etc.)

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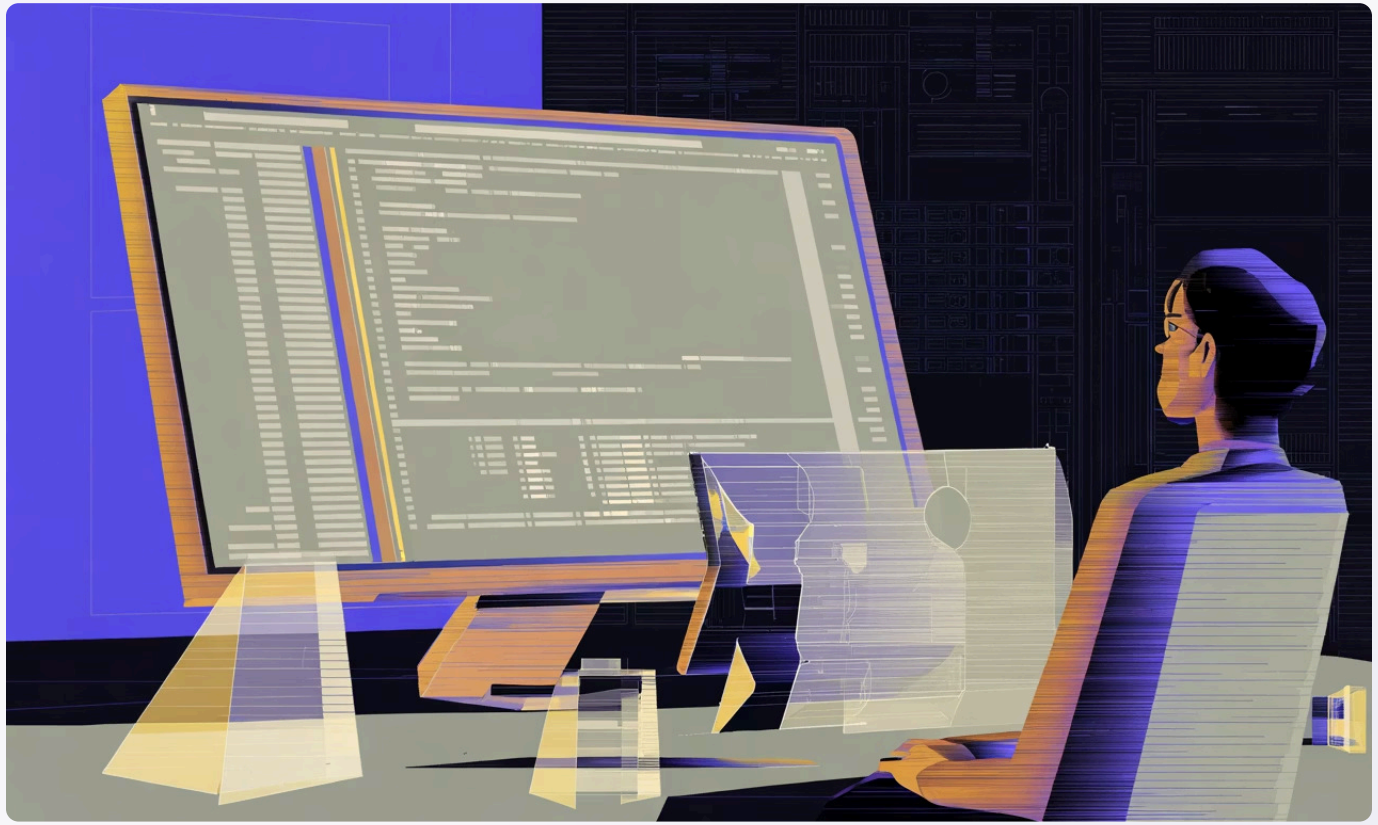
EDA: Data Loading & Initial Exploration



What Was Done

- Loaded dataset with `pd.read_csv()`
- Explored structure via df.head(), `df.tail()`, df.info(), df.describe()
- Checked missing values with `df.isnull().sum()`

Why It Matters

Early identification of data quality issues ensures accurate, reliable downstream analysis.

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Data Cleaning & Preparation

Category Correction

Built a mapping dictionary to fix inconsistencies between Item Purchased and Category — critical for accurate grouping.

Missing Values — Smart Imputation

Size: "Not Applicable" for Electronics/Accessories, "Free Size" for Footwear, mode for Clothing. Review Rating: product-level mean. Purchase Amount: mean. Previous Purchases: 0.

Duplicates & Standardization

Removed duplicate Customer IDs. Standardized column names (lowercase, underscores) for SQL and Power BI compatibility.

Export to SQL Server

Used `pyodbc` and `sqlalchemy` to load cleaned data into SQL for query-based analysis and BI integration.

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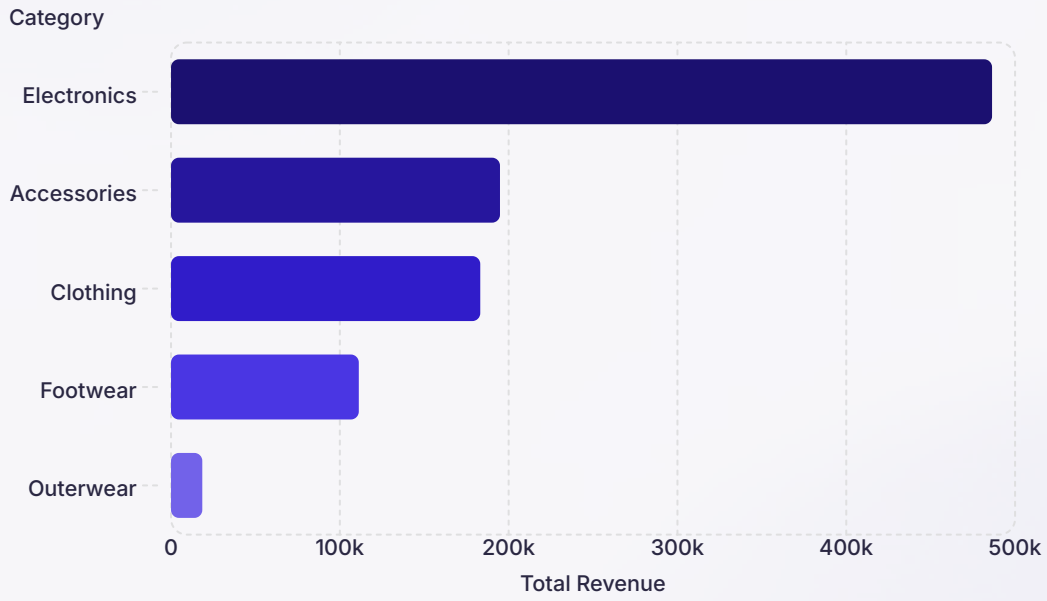
SQL Analysis: Key Findings



Top Insights

- **Electronics** leads revenue at $486K
- Discounts raise avg. purchase from **$182 → $219**
- Subscribers spend **$270.58** avg vs. **$165.72** non-subscribers
- Loyal Customers: **3,085** | Returning: **1,354** | New: **561**
- Age 51+ contributes highest revenue: **$399K (40.17%)**

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Power BI Dashboard Overview

Interactive dashboard with filters for Category, Gender, Discount, and Subscription Status.

$994K

Total Spend

198.84

Avg Spend

5K

Total Customers

30

Total Items

3.62

Avg Rating

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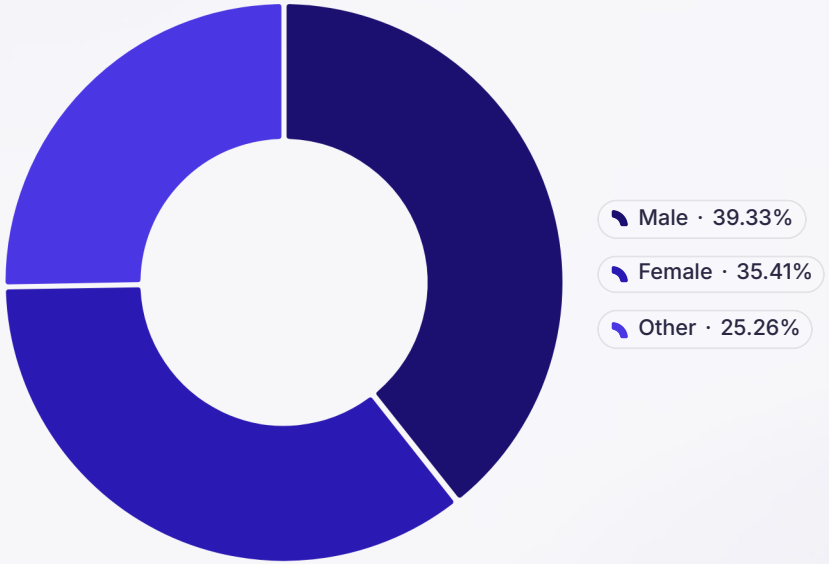
Dashboard Visuals: Revenue Breakdown



Revenue By Gender

- Male: **$390.96K (39.33%)**
- Female: **$352.06K (35.41%)**
- Other: **$251.16K (25.26%)**

Revenue By Age Group

- 51+: **$399.35K (40.17%)**
- 35–50: **$266.71K (26.83%)**
- 26–35: **$177.42K (17.85%)**
- 18–25: **$150.71K (15.16%)**

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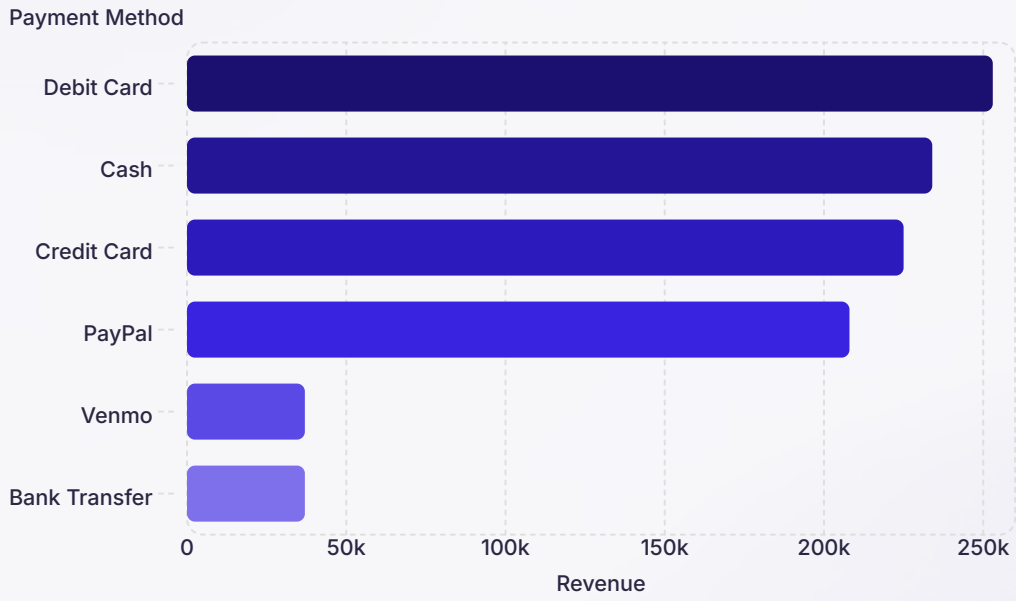
Dashboard Visuals: Products, Payments & Locations



Top 5 Items By Revenue

- Headphones: **$138K**
- Phone: **$125.6K**
- Bag: **$120.7K**
- Laptop: **$115.2K**
- Watch: **$107.6K**

Top Locations By Revenue

- New York: **$187.12K**
- Los Angeles: **$159.99K**
- Phoenix: **$142.40K**

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